

NAWAL SHAHID

Full Stack Engineer | React.js • Node.js • MERN Stack
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SUMMARY

Full stack engineer with 3+ years of hands-on experience building and deploying production-grade web and mobile applications using the MERN stack (MongoDB, Express.js, React.js, Node.js). Delivered an accessibility-first ed-tech platform achieving WCAG 2.1 AA compliance. Built ML classifiers with 93.1% accuracy across 40,000+ article datasets. Hands-on experience across REST API design, JWT and OAuth2 authentication, role-based access control, Firebase serverless architecture, Apache Kafka event-driven streaming, Docker containerization, and CI/CD deployment pipelines. Deployed REST APIs behind Nginx reverse proxy with environment-based configuration, MongoDB indexing strategy, and production-ready error handling.

EXPERIENCE

Full Stack Web and Mobile Developer | [Kazim Trust — DigiLex Ed-Tech Platform](#) | Karachi, PK | 2024 – 2026

- Architected the complete MERN stack platform for DigiLex, designing a Node.js/Express.js REST API with JWT authentication, role-based access control (Admin/Learner), and MongoDB schema optimized with compound indexes for content retrieval queries, reducing average API response time from approximately 420ms to 180ms.
- Built the React.js frontend with Tailwind CSS using a reusable component library, achieving a Lighthouse accessibility score of 93 and full WCAG 2.1 AA compliance, verified against screen readers and assistive devices in collaboration with occupational therapists at Kazim Trust.
- Developed the cross-platform DigiLex mobile app in React Native with Expo, integrating Firebase Authentication and Firestore with offline persistence, delivering consistent performance across Android and iOS and reducing session drop-off by approximately 40% versus the prior web-only flow.
- Improved Time to Interactive by an estimated 38% on low-bandwidth connections by implementing React lazy loading, route-level code splitting, and client-side caching with service workers.

Web Developer Apprentice | [AJD Marketing](#) | Qatar (Remote) | 2023 – 2024

- Built and maintained 4 responsive client websites using React.js, PHP, MySQL, and Tailwind CSS, owning the full stack from database schema design to frontend component architecture and production deployment on Netlify.
- Implemented SEO optimizations including semantic HTML5, JSON-LD structured data, and Core Web Vitals tuning, improving average Lighthouse performance score from 68 to 91 and reducing average page load time from 3.4s to 1.7s across client sites.
- Refactored legacy PHP backend and SQL query layer, replacing N+1 query patterns with JOIN-optimized queries and introducing query result caching, cutting server response time by approximately 45% on high-traffic pages.

PROJECTS

NexaPress — Full Stack CMS (MERN Version) | MongoDB | Express.js | React.js | Node.js | JWT | REST API | Vercel

- Designed and built a Medium-style content management system with admin-only publishing and a public read interface, implementing JWT authentication, role-based authorization, and a structured MongoDB schema with categories, tags, SEO metadata fields, and dynamic routing.
- Deployed full-stack application on Vercel with environment-based configuration, secure API key handling, and a scalable backend structure ready for analytics integration and multi-author workflows.

NexaPress — Serverless CMS (Firebase Version) | React | Firebase Auth | Firestore | Cloud Functions | Security Rules

- Rebuilt the CMS on a serverless Firebase architecture with real-time Firestore content synchronization, enforcing write restrictions via granular Firestore security rules and Firebase Authentication with a protected admin dashboard.
- Automated backend logic using Cloud Functions for content publication workflows, achieving rapid deployment with near-zero infrastructure maintenance overhead.

Enterprise Security Streaming and Threat Analytics Platform | Apache Kafka | Python | Docker | Event-Driven Architecture | Stream Processing

- Designed an event-driven streaming pipeline using Apache Kafka with a multi-partition topic strategy (8 partitions per topic) and dedicated consumer groups, supporting throughput of approximately 50,000 log events per minute with fault-tolerant offset management and automatic consumer rebalancing.
- Built Python stream processing consumers for real-time anomaly detection with configurable alert thresholds, containerized all services using Docker Compose for modular and reproducible local and staging deployment.

News Classifier — ML Article Classification System Python | scikit-learn | Streamlit | TextBlob | BeautifulSoup

- Achieved 93.1% classification accuracy across 5 news categories on a 40,000+ article dataset using an ensemble of Naive Bayes, SVM, and Random Forest, with real-time sentiment analysis and per-prediction explainability (feature importance, class probability).
- Deployed an interactive Streamlit dashboard supporting URL-based input, live batch upload, word cloud visualization, and downloadable CSV prediction exports.

TECHNICAL SKILLS

Frontend: React.js, Next.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS, Material UI, WCAG 2.1, Responsive Design

Backend and APIs: Node.js, Express.js, REST API Design, JWT, OAuth2, Role-Based Access Control, Cloud Functions, Serverless Architecture, Nginx

Databases: MongoDB, Mongoose, Firebase Firestore, PostgreSQL, MySQL, Schema Design, Query Optimization, Compound Indexing

Mobile: React Native, Expo, Android Studio, Kotlin, Firebase Auth, Firestore, Offline Persistence, Cross-Platform Deployment

Streaming and DevOps: Apache Kafka, Event-Driven Architecture, Docker, Docker Compose, GitHub Actions, CI/CD, Git, Netlify, Vercel, Postman

Data and ML: Python, scikit-learn, Pandas, TextBlob, BeautifulSoup, Streamlit, Ensemble Models, Sentiment Analysis

EDUCATION

BSc Software Engineering | Jinnah University for Women, Karachi | 2022 – 2026 | CGPA: 3.5 / 4.0

Technical Management Co-Lead, Google Developer Student Club (GDSC) | Creative Art Manager, University Event Branding